

# ELEC4631 Lectures 3 and 4: Dynamical Systems and Linear State- Space Systems

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# Outline

- State-space, states, and dynamical systems.
- Continuous-time dynamical systems.
- Linear and linear time-invariant state-space systems

# Dynamical systems

- Systems that are evolving in time and involve time-varying variables are ubiquitous In various branches of engineering,
- Such systems often referred to as *dynamical systems*.

# Dynamical systems

- Whether system is electric circuit, car, aircraft, power system or chemical plant, often their dynamics are described by a set of equations that share some common properties.
- Thus many of these systems, although belonging to very different physical domains, can be treated with a common mathematical framework.

# State-space

- First need to formally define dynamical systems. To do this, let's begin with notion of a state-space.
- A *state-space* is a set of coordinates that is sufficient (in a mathematical model) to give a complete description of the dynamics of a system of interest.
- Elements of state-space referred to as *states*.

# State-space

- Example 1: Linear RLC circuit.

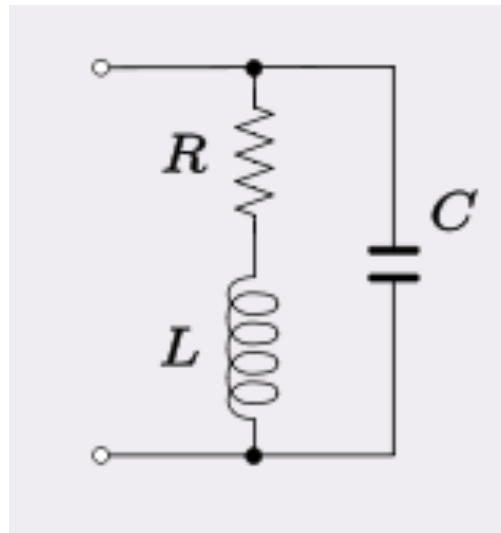


Image source: Wikipedia

- In absence of driving voltage or current source, dynamics completely described by time varying electric field  $E(t)$  across capacitor and magnetic field  $B(t)$  across inductor.

# State-space

- Thus a candidate state-space for RLC circuit is a set containing all pairs  $(e, b)$  of all possible values  $e$  and  $b$  that  $E(t)$  and  $B(t)$  can take for any  $t$ , respectively.
- However,  $E(t)$  depends on voltage  $v_C(t)$  across the capacitor and  $B(t)$  depends on current passing through inductor  $i_L(t)$ .
- Hence, another candidate state-space for circuit is a set containing all pairs  $(v, i)$  of all possible values  $v$  and  $i$  that  $v_C(t)$  and  $i_L(t)$  can take for any  $t$ , respectively.

# State-space

- Example 2: Pendulum. Motion of pendulum can be completely described by angle  $\theta(t)$  and angular velocity  $\omega(t)$ .
- Thus, a candidate state-space for pendulum is a set that contains all pairs  $(A, B)$  of all possible values  $A$  of angle  $\theta(t)$  and all possible values  $B$  of angular velocity  $\omega(t)$ .

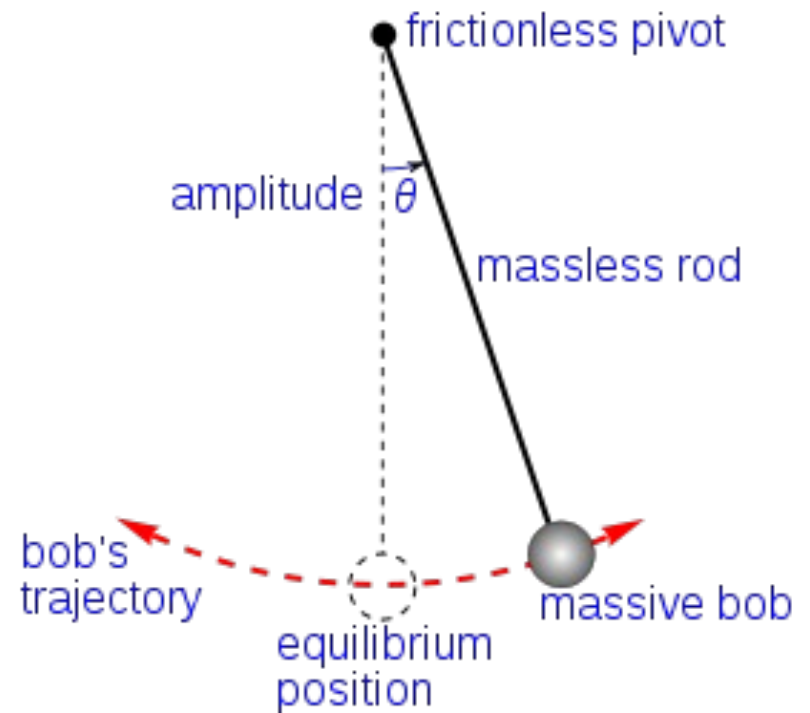


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# Dynamical system

- Roughly speaking, a *dynamical system* is a *rule* for *evolution* on a state-space  $S$ .
- Rule dictates how the state will move in  $S$  over time.
- Time can be discrete or continuous, depending on system.

# Continuous-time dynamical systems

- In this course our interest is with *continuous-time dynamical systems*.
- Time is continuous and  $S$  is Euclidean space.
- Rule is given by *ordinary differential equations (ODEs)*.
- Rule produces curves traced out by the states over time on state-space, called *trajectories*.

# Continuous-time dynamical systems

- Trajectory  $x$  (a function of time) satisfies ODE

$$\dot{x}(t) = f(x(t)); x(0) = x_0$$

where  $x_0$  is initial state at  $t = 0$ .

(Recall that a dot on top denotes differentiation with respect to time).

- $f(x)$  is the instantaneous velocity of the trajectory at a state  $x$ . It is a vector for each  $x$ .
- The collection of all vectors  $f(x)$  for all  $x$  form a *vector field*. Thus  $f$  referred to as vector field.

# Visualising the vector field $f$

- For system with  $S = \mathbb{R}^2$  and  $f(x,y) = (-ay, ax)$ ,  $a > 0$ , this is what  $f$  looks like:

$f(x,y)$  at any point  $(x,y)$  in  $S$  gives instantaneous velocity of dynamical system at that point (indicated by a vector arrow). Length of arrow proportional to magnitude of velocity at that point.

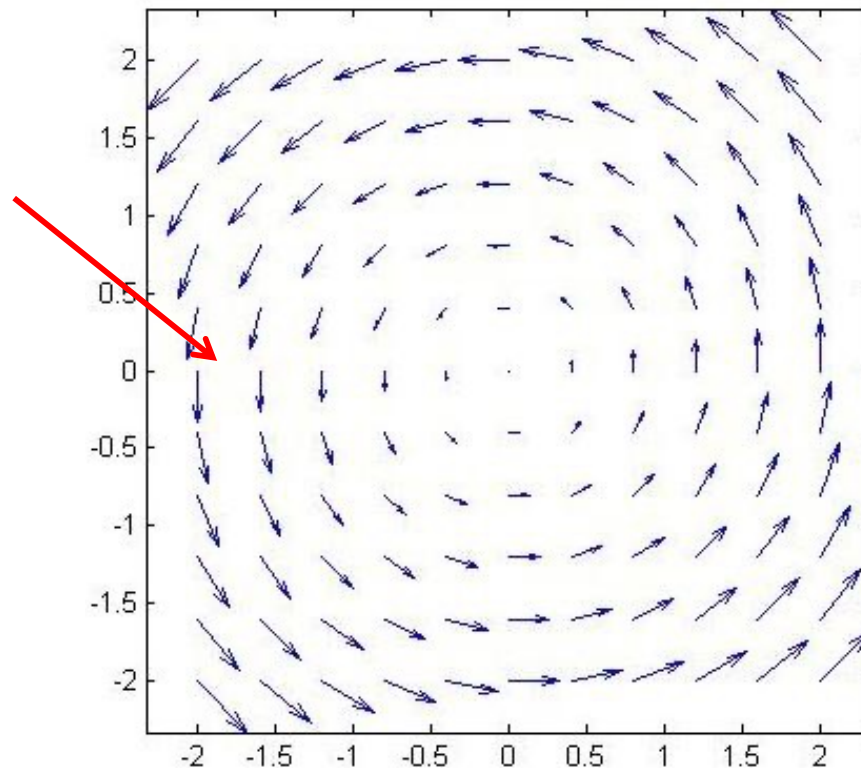


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[quasirandomideas.wordpress.com](http://quasirandomideas.wordpress.com)

# RLC circuit

- Example: RLC circuit.

State is  $(v_C, i)$ .

$$\dot{v}_C = \frac{dv_C}{dt} = \frac{1}{C}i, \quad v_C(0) = v_0$$

$$\dot{i} = \frac{di}{dt} = -\frac{1}{L}v_C - \frac{R}{L}i, \quad i(0) = i_0$$

where  $(v_0, i_0)$  is initial state.

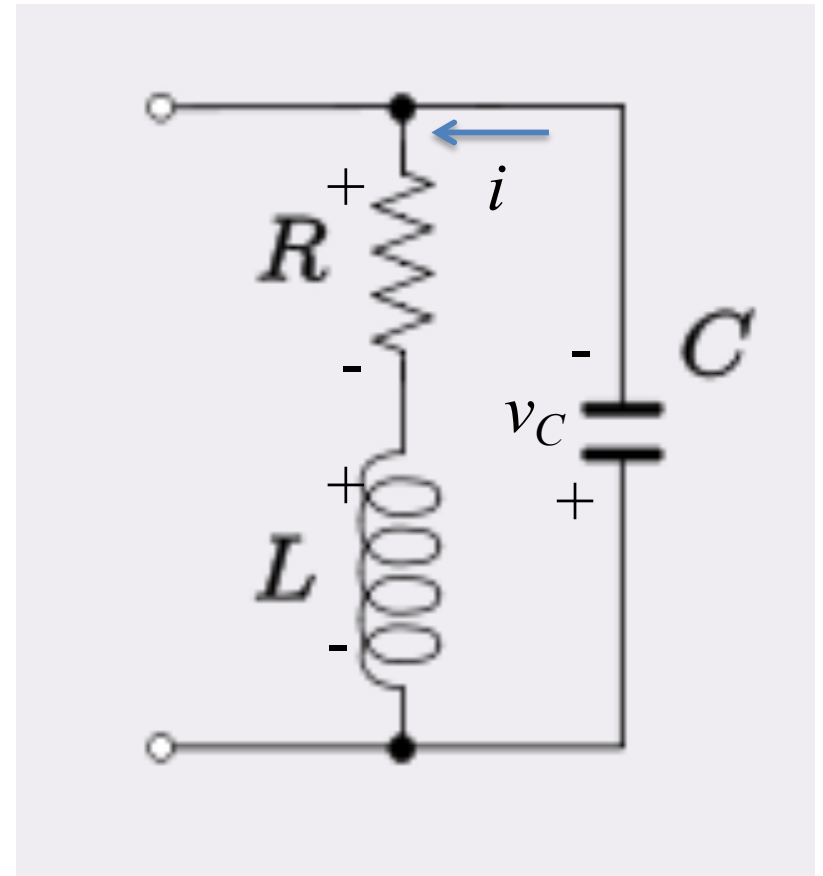


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# RLC circuit

- We can take state-space for RLC circuit to be  $S = \mathbb{R}^2$
- So, for RLC circuit

$$f(v_C, i) = \begin{bmatrix} \frac{1}{C}i \\ -\frac{1}{L}v_C - \frac{R}{L}i \end{bmatrix}$$

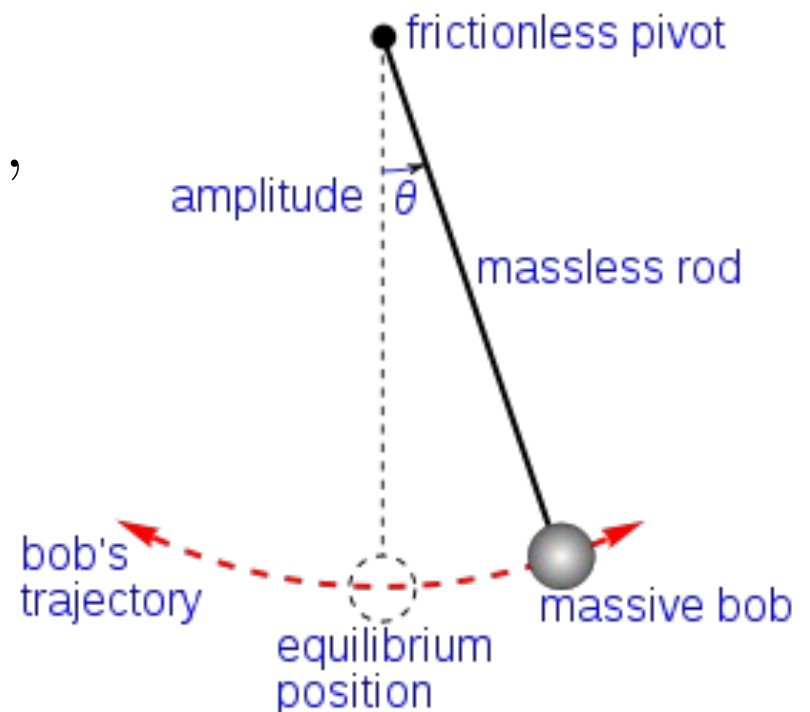
# Pendulum

- Example: Pendulum. Laws of physics give equation of motion for the pendulum by pair of coupled ODEs

$$\dot{\theta}(t) = \omega(t), \theta(0) = \theta_0$$

$$\dot{\omega}(t) = -\frac{g}{r} \sin(\theta(t)), \omega(0) = \omega_0,$$

where  $(\theta_0, \omega_0)$  is initial state,  
 $r$  radius of pendulum, and  $g$   
gravitational acceleration  
constant.



# Pendulum

- We can take state-space of pendulum to be  $S = \mathbb{R}^2$
- So, for pendulum

$$f(\theta, \omega) = \begin{bmatrix} \omega \\ -\frac{g}{r} \sin(\theta) \end{bmatrix}$$



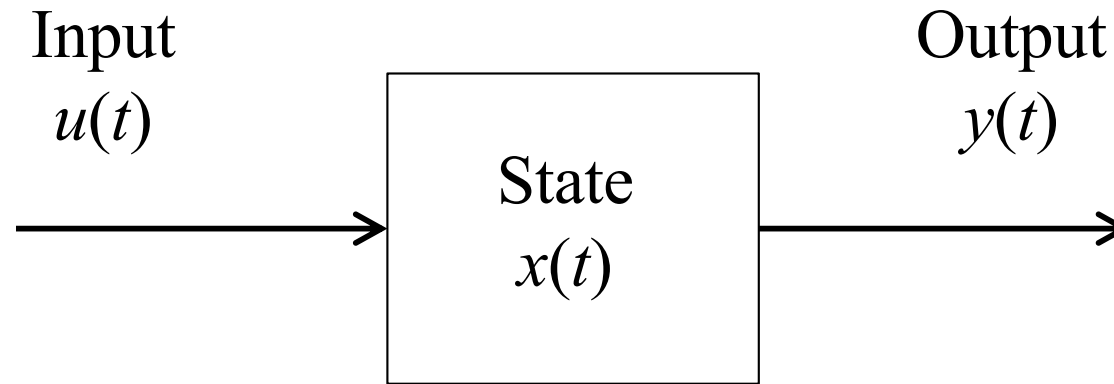
# Controlled dynamical systems

- More generally, there are dynamical systems that can be externally controlled through a control signal  $u$
- The signal  $u$  can be used to influence (control) the behaviour of the system from the outside, to force the system to behave in a desired way
- The system can also have an output signal  $y$ . This is a signal that is produced by the system and can be observed or measured externally

# Controlled dynamical systems

- There can in general be multiple inputs and multiple outputs, where  $u(t)$  in  $\mathbb{R}^n$  and  $y(t)$  in  $\mathbb{R}^p$ , with  $n \geq 1$  and  $p \geq 1$

# Controlled dynamical systems



- The input signal  $u$  and output signal  $y$  can be accessed externally (from outside the system)
- The state  $x$  may not be externally observable

# Controlled dynamical systems

- A controlled dynamical system is described by a controlled ODE of the form:

$$\dot{x}(t) = f(x(t), u(t))$$

where  $f$  is a controlled vector field,  $f: \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$

# Controlled dynamical systems

- The output can take the general form

$$y(t) = h(x(t), u(t))$$

where  $h : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^p$

- If the state  $x$  can be externally observed then often the output  $y$  is taken to be  $y(t) = x(t)$

# Linear systems

- Linear systems are a special and basic type of controlled dynamical system that is very important in control systems
- They are important because many physical systems can be approximately modelled as linear systems
- In these systems there is a *linear* relationship between the *input*, *state* and *output*

# Input-output models

- In ELEC3114 (or an equivalent course) you have encountered input-output systems represented by differential equations:

$$y^{(n)}(t) + a_{n-1}y^{(n-1)}(t) + \dots + a_0y(t) = b_m u^{(m)}(t) + b_{m-1}u^{(m-1)}(t) + \dots + b_0u(t)$$

with  $n \geq m$ ,  $u(t)$  the input, and  $y(t)$  the output.

- Taking Laplace transform of  $u$  and  $y$  gives the transfer function

$$H(s) = \frac{Y(s)}{U(s)} = \frac{b_ms^m + b_{m-1}s^{m-1} + \dots + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_0}$$

# However ...

- All input-output systems represented by linear differential equations or transfer functions can be transformed into state-space form.
- State-space formalism has proven to be a very powerful tool.
- A number of difficult problems in control theory was solved with state space approach. For instance, the so-called  $H^\infty$  control problem for robust control.



# First example of conversion

- For *small* initial conditions,  $\sin \theta(t) \approx \theta(t)$  for all  $t \geq 0$  and  $\theta(t)$  in the undamped pendulum example *approximately* satisfies the linear ODE:

$$mr\ddot{\theta}(t) + mg\theta(t) = 0, \quad (\theta(0), \dot{\theta}(0)) = (\theta_0, \dot{\theta}_0)$$

- Introduce now state variable  $x(t) = (x_1(t), x_2(t))^T$  with

$$x_1(t) = \theta(t),$$

$$x_2(t) = \dot{\theta}(t).$$

# First example of conversion

- Then we have

$$\dot{x}_1(t) = \dot{\theta}(t) = x_2(t),$$

$$\dot{x}_2(t) = \ddot{\theta}(t) = -\frac{g}{r}\theta(t) = -\frac{g}{r}x_1(t)$$

- So we get a linear state-space representation of the undamped pendulum as

$$\dot{x}(t) = \begin{bmatrix} x_2(t) \\ -\frac{g}{r}x_1(t) \end{bmatrix}, \quad x(0) = \begin{bmatrix} \theta(0) \\ \dot{\theta}(0) \end{bmatrix}$$

# Some history

- State-space representation became integral element of control theory through influential work of Rudolf Kalman in the late 50s and in the 60s.
- It brings in and exploits the idea of the state-space and state from dynamical systems theory (Lecture 3).
- Before Kalman's work, control theory up to the 50s relied mostly on complex analysis and complex function theory (e.g., Laplace transforms). Work up to the 50's often known as *classical control theory*.

# Some history

- Kalman's contributions brought linear algebra, optimisation theory, and the calculus of variations as important additional tools for control theory in addition to complex analysis and function theory.
- This gave birth to new era in control theory from 1960s onwards known as era of *modern control theory*.

# Linear state-space systems

- A single input-single output (SISO) linear state-space system with scalar input  $u(t)$  and scalar output  $y(t)$  has general form:

$$\dot{x}(t) = A(t)x(t) + B(t)u(t); x(0) = x_0$$

$$y(t) = C(t)x(t) + D(t)u(t)$$

$$x(t) \in R^n, A(t) \in R^{n \times n}, B(t) \in R^{n \times 1}, C(t) \in R^{1 \times n}, D(t) \in R$$

- $x(t)$  is the *state* of the system and  $n$  is called the *order* of the system. Also,  $x_0$  is the initial state.

# Linearity

- Let  $x_j(t)$  and  $y_j(t)$  satisfy ODE with input  $u_j(t)$  and initial condition  $x_j(0) = x_{j0}$  for  $j = 1, 2$ :

$$\dot{x}_j(t) = A(t)x_j(t) + B(t)u_j(t); x_j(0) = x_{j0}$$

$$y_j(t) = C(t)x_j(t) + D(t)u_j(t)$$

# Linearity

- Then for any real  $c_1$  and  $c_2$ ,  $c_1x_1(t) + c_2x_2(t)$  and  $c_1y_1(t) + c_2y_2(t)$  satisfy the ODE with input  $c_1u_1(t) + c_2u_2(t)$  and initial condition  $c_1x_1(0) + c_2x_2(0) = c_1x_{10} + c_2x_{20}$ :

$$c_1\dot{x}_1(t) + c_2\dot{x}_2(t) = A(t)(c_1x_1(t) + c_2x_2(t)) + B(t)(c_1u_1(t) + c_2u_2(t));$$

$$c_1x_1(0) + c_2x_2(0) = c_1x_{10} + c_2x_{20},$$

$$c_1y_1(t) + c_2y_2(t) = C(t)(c_1x_1(t) + c_2x_2(t)) + D(t)(c_1u_1(t) + c_2u_2(t))$$

# Linearity

- That is, linearity is with respect to the quartet  $(x_0, u, x, y)$ .
- If  $(x_{10}, u_1, x_1, y_1)$  is one quartet satisfying ODE and  $(x_{20}, u_2, x_2, y_2)$  is another quartet satisfying ODE then the quartet  $(c_1x_{10}+c_2x_{20}, c_1u_1+c_2u_2, c_1x_1+c_2x_2, c_1y_1+c_2y_2)$  also satisfies ODE for any real numbers  $c_1$  and  $c_2$ .



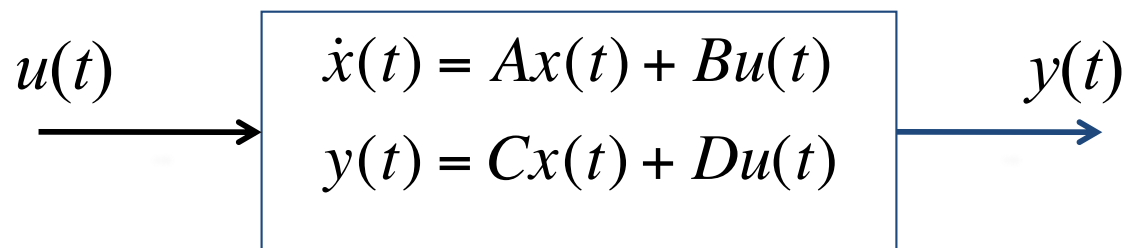
# Linear time-invariant (LTI) state-space representation

- When  $A(t)$ ,  $B(t)$ ,  $C(t)$  and  $D(t)$  independent of  $t$  then SISO state-space system said to be *linear time-invariant* (LTI) and has the form:

$$\dot{x}(t) = Ax(t) + Bu(t); x(0) = x_0$$

$$y(t) = Cx(t) + Du(t),$$

$$t \geq 0, x(t) \in R^n, A \in R^{n \times n}, B \in R^{n \times 1}, C \in R^{1 \times n}, D \in R$$



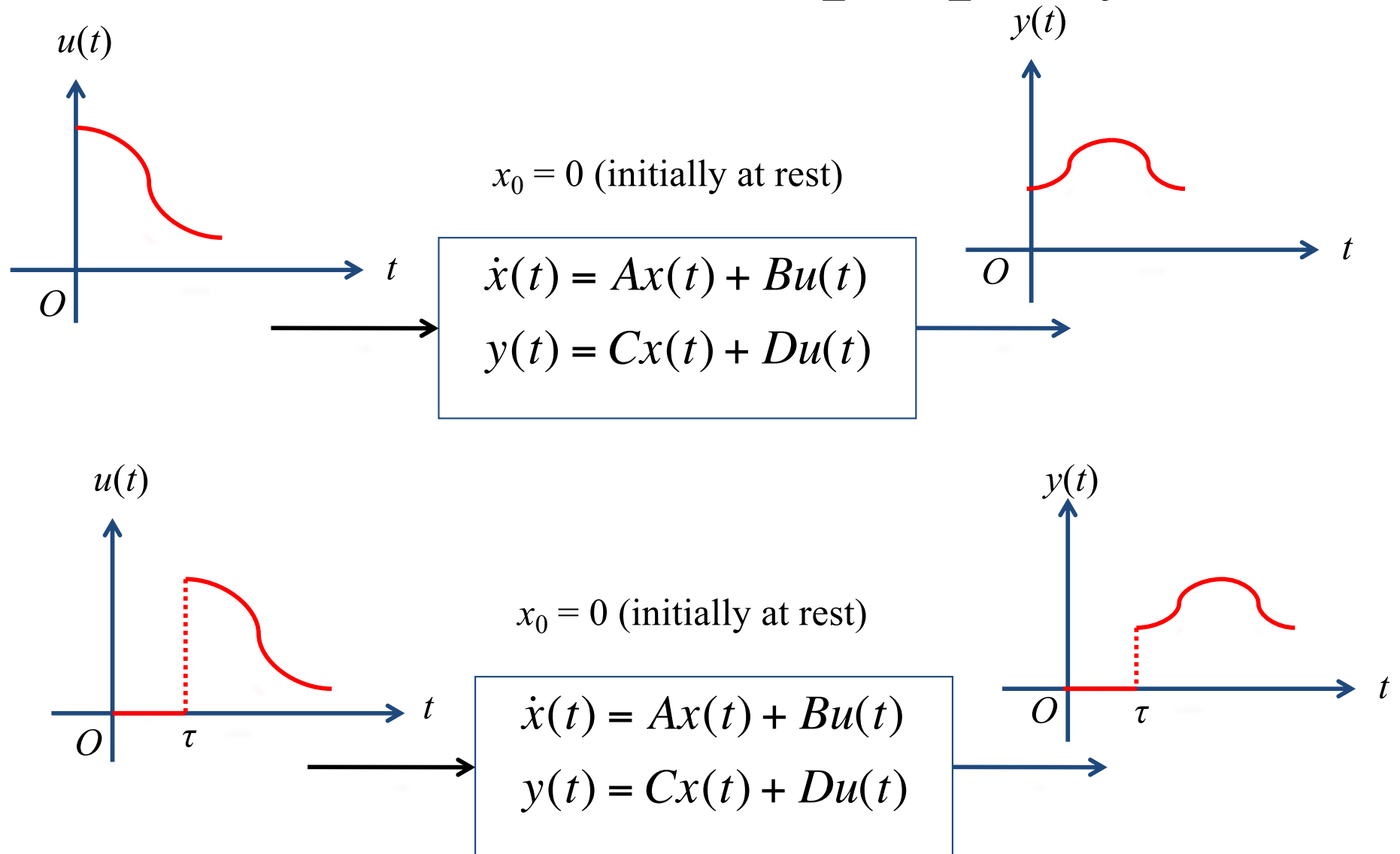
# Time-invariance property

- Consider an LTI system with initial condition  $x_0 = 0$  with input  $u(t)$  and output  $y(t)$ .
- Let  $1(t)$  denote step function

$$1(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$$

- Then if  $(u(t), y(t))$  is input-output pair of initially at rest LTI system (i.e.,  $x_0 = 0$ ), so is  $(u(t-\tau)1(t-\tau), y(t-\tau)1(t-\tau))$  for any constant  $\tau > 0$ .

# Time-invariance property



# Remarks

- This course will largely focus on SISO linear state-space systems.
- However, some elements of multiple-input multiple-output systems will make an appearance later on.

# Transfer function from state-space representation

- Consider again the (LTI) state-space equation

$$\dot{x}(t) = Ax(t) + Bu(t); x(0) = x_0$$

$$y(t) = Cx(t) + Du(t)$$

- Taking  $x_0 = 0$  (initially at rest system) and Laplace transforms on both sides of equation gives

$$\begin{aligned} & \Leftrightarrow \begin{aligned} (sI - A)X(s) &= BU(s) \\ Y(s) &= CX(s) + DU(s) \end{aligned} \\ & \Leftrightarrow \begin{aligned} X(s) &= (sI - A)^{-1}U(s) \\ Y(s) &= (C(sI - A)^{-1}B + D)U(s) \end{aligned} \\ & \Leftrightarrow \boxed{\frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D} \end{aligned}$$

# Example of transfer function calculation

- Consider state-space system

$$\dot{x}(t) = \begin{bmatrix} -2 & 1 \\ 3 & 1 \end{bmatrix} x(t) + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u(t); \quad x(0) = x_0$$

$$y(t) = \begin{bmatrix} 1 & -4 \end{bmatrix} x(t) + 2u(t)$$

# Example of transfer function calculation

- Then

$$\begin{aligned} X(s) &= (sI_{2 \times 2} - A)^{-1} BU(s) \\ &= \left( sI_{2 \times 2} - \begin{bmatrix} -2 & 1 \\ 3 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} -1 \\ 1 \end{bmatrix} U(s) \\ &= \left( \begin{bmatrix} s+2 & -1 \\ -3 & s-1 \end{bmatrix} \right)^{-1} \begin{bmatrix} -1 \\ 1 \end{bmatrix} U(s) \\ &= \frac{1}{s^2 + s - 5} \begin{bmatrix} -s+2 \\ s-1 \end{bmatrix} U(s) \\ &= \begin{bmatrix} \frac{-s+2}{s^2 + s - 5} U(s) \\ \frac{s-1}{s^2 + s - 5} U(s) \end{bmatrix} \end{aligned}$$

# Example of transfer function calculation

- Therefore,

$$\begin{aligned} H(s) &= \frac{Y(s)}{U(s)} \\ &= \frac{\begin{bmatrix} 1 & -4 \end{bmatrix} X(s) + 2U(s)}{U(s)} \\ &= \frac{-5s + 6 + 2(s^2 + s - 5)}{s^2 + s - 5} \\ &= \frac{2s^2 - 3s - 4}{s^2 + s - 5} \end{aligned}$$



# Conversion from transfer function to state-space representation

- Suppose we have transfer function

$$H(s) = \frac{Y(s)}{U(s)} = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_0},$$

with  $m < n$  (the case  $m = n$  discussed in next slide). How can we find state-space system with this transfer function?

- We'll look at three state-space forms: *control canonical form*, *modal canonical form*, and *observer canonical form*.

# General principle for $n = m$

- Once the case  $m < n$  sorted out, easy to handle the case  $m = n$ . Simply write

$$\begin{aligned}\frac{Y(s)}{U(s)} &= \frac{b_n s^n + b_{n-1} s^{n-1} + \dots + b_0}{s^n + a_{n-1} s^{n-1} + \dots + a_0}, \quad b_n \neq 0 \\ &= b_n + \frac{(b_{n-1} - b_n a_{n-1}) s^{n-1} + (b_{n-2} - b_n a_{n-2}) s^{n-2} + \dots + b_0 - b_n a_0}{s^n + a_{n-1} s^{n-1} + \dots + a_0}\end{aligned}$$

Numerator has degree less than  $n$ . Put this part into state space form!

- Output will be of form

$$y(t) = \text{some linear function of state } x(t) + b_n u(t)$$

# Control canonical form

- Consider transfer function

$$H(s) = \frac{Y(s)}{U(s)} = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_0},$$

and introduce  $Z(s)$  to split  $H(s)$  into two factors:

$$\frac{Z(s)}{U(s)} = \frac{1}{s^n + a_{n-1}s^{n-1} + \dots + a_0}$$

$$\frac{Y(s)}{Z(s)} = b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_0$$

# Control canonical form

- First consider

$$\frac{Z(s)}{U(s)} = \frac{1}{s^n + a_{n-1}s^{n-1} + \dots + a_0}.$$

- Let  $X_1(s) = Z(s)$ ,  $X_2(s) = sZ(s)$ , ...,  $X_n(s) = s^{n-1}Z(s)$ .
- Then  $sX_1(s) = X_2(s)$ ,  $sX_2(s) = X_3(s)$ , ...,  $sX_{n-1}(s) = X_n(s)$  and

$$sX_n(s) = -a_{n-1}X_n(s) - a_{n-2}X_{n-1}(s) - \dots - a_0X_1(s) + U(s)$$

# The control canonical form

- Let  $x_1(t), x_2(t), \dots, x_n(t)$  be inverse Laplace transform of  $X_1(s), X_2(s), \dots, X_n(s)$ , resp., with  $x_1(0) = x_2(0) = \dots = x_n(0) = 0$ .
- Then by inverse Laplace transform, equations equivalent to:

$$\dot{x}_1(t) = x_2(t)$$

$$\dot{x}_2(t) = x_3(t)$$

$$\vdots$$

$$\dot{x}_{n-1}(t) = x_n(t)$$

$$\dot{x}_n(t) = -a_{n-1}x_n(t) - a_{n-2}x_{n-1}(t) - \dots - a_0x_1(t) + u(t)$$

# The control canonical form

- Collecting things together:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \vdots \\ \dot{x}_{n-1}(t) \\ \dot{x}_n(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_{n-1}(t) \\ x_n(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u(t)$$

# The control canonical form

- We now turn to

$$\frac{Y(s)}{Z(s)} = b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_0$$

$$Y(s) = b_{n-1}X_n(s) + b_{n-2}X_{n-1}(s) + \dots + b_0X_1(s)$$

$$y(t) = b_{n-1}x_n(t) + b_{n-2}x_{n-1}(t) + \dots + b_0x_1(t)$$

which by Laplace transform inversion gives

# The control canonical form

- Collecting things together again:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \vdots \\ \dot{x}_{n-1}(t) \\ \dot{x}_n(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_{n-1}(t) \\ x_n(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} b_0 & b_1 & \dots & b_{n-1} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_{n-1}(t) \\ x_n(t) \end{bmatrix}$$



# The control canonical form

- Writing

$$x(t) = (x_1(t), x_2(t), \dots, x_n(t))^T$$

then

Control canonical  
form when  $m < n$



$$\dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-1} \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u(t)$$
$$y(t) = [b_0 \quad b_1 \quad \dots \quad b_{n-1}] x(t)$$

If  $m = n$  then in the control canonical case ...

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-1} \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u(t)$$

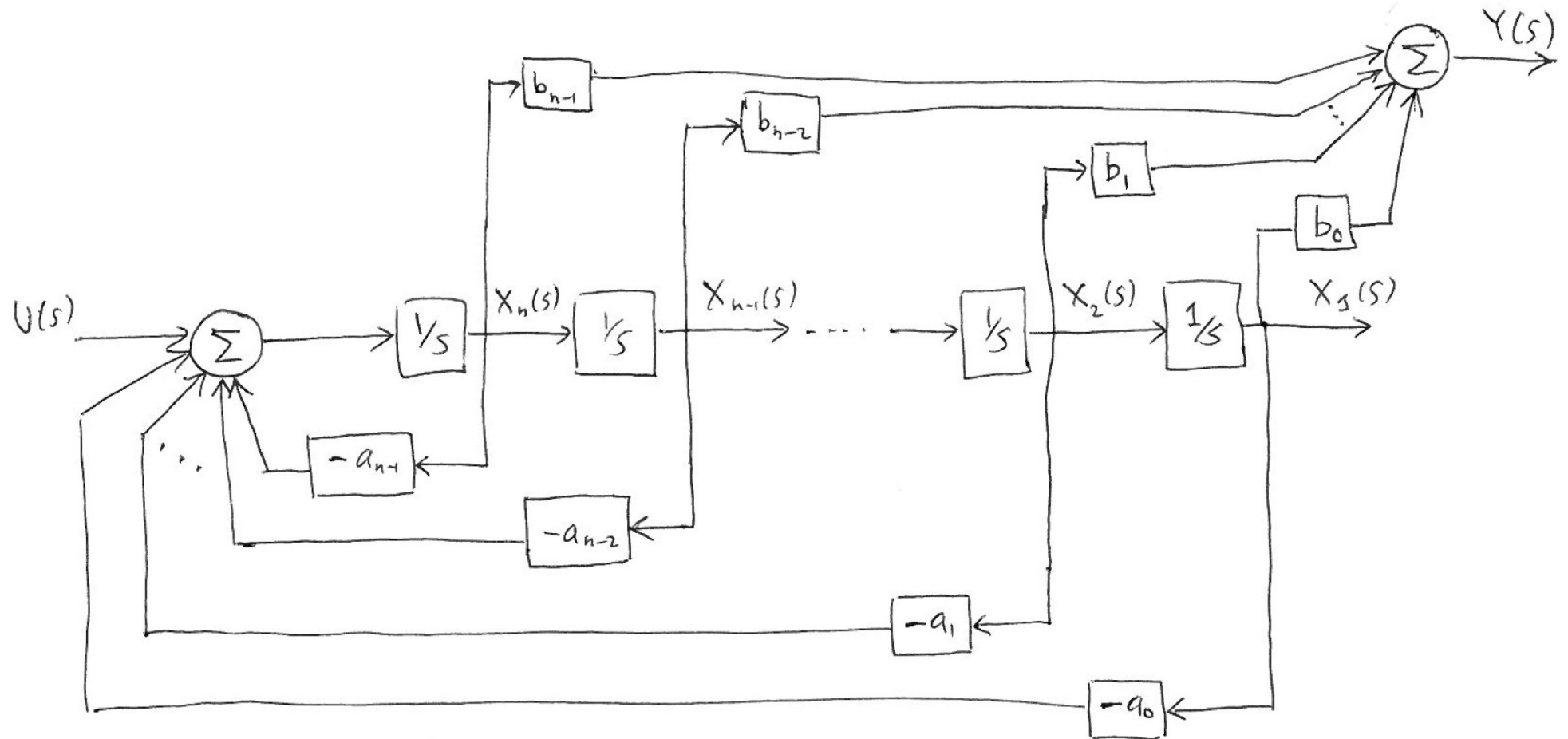
$$y(t) = [b_0 - b_n a_0 \quad b_1 - b_n a_1 \quad \dots \quad b_{n-1} - b_n a_{n-1}] x(t) + b_n u(t)$$

$$Y_1(s) = \frac{(b_{n-1} - b_n a_{n-1})s^{n-1} + (b_{n-2} - b_n a_{n-2})s^{n-2} + \dots + b_0 - b_n a_0}{s^n + a_{n-1}s^{n-1} + \dots + a_0} U(s)$$

$$Y_2(s) = b_n U(s)$$

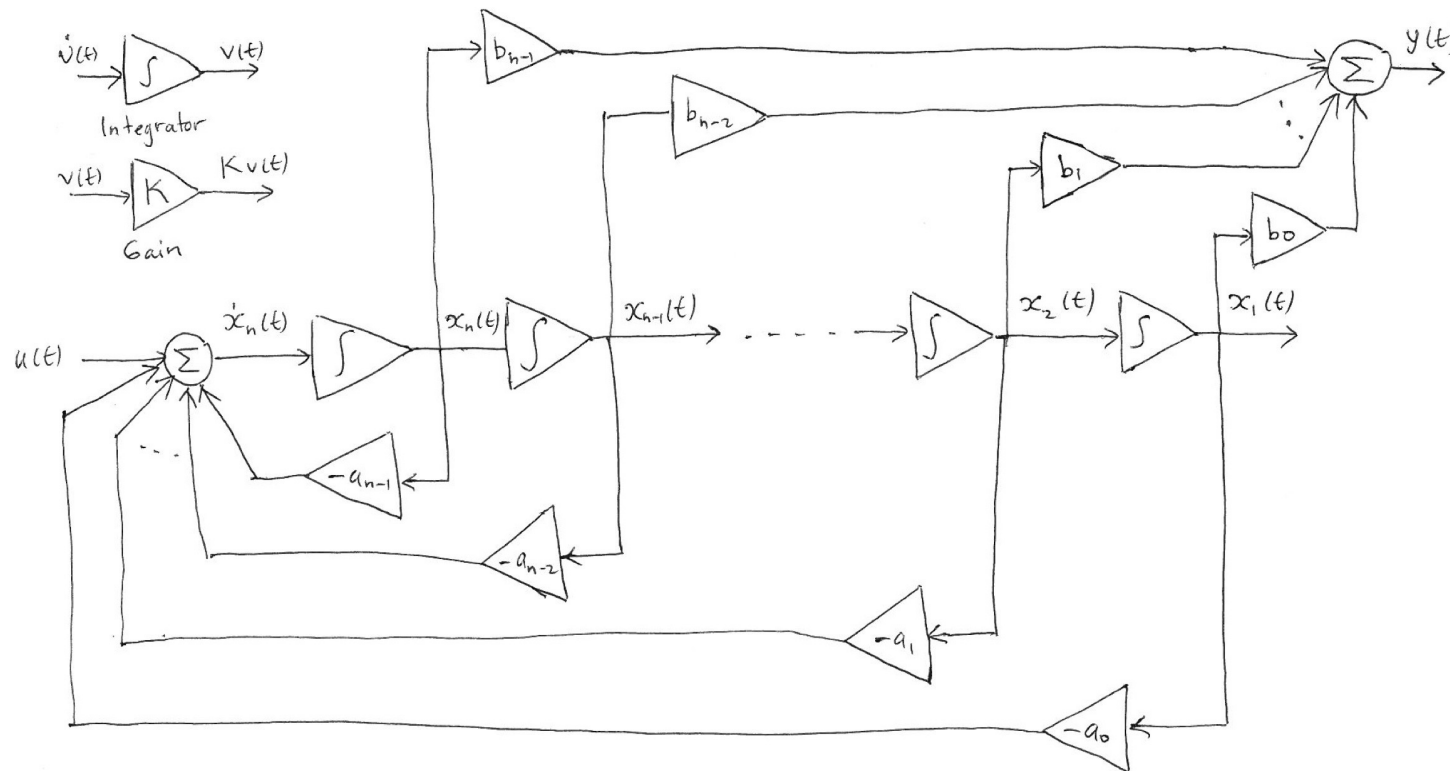
$$Y(s) = Y_1(s) + Y_2(s)$$

# Control canonical block diagram ( $m < n$ )



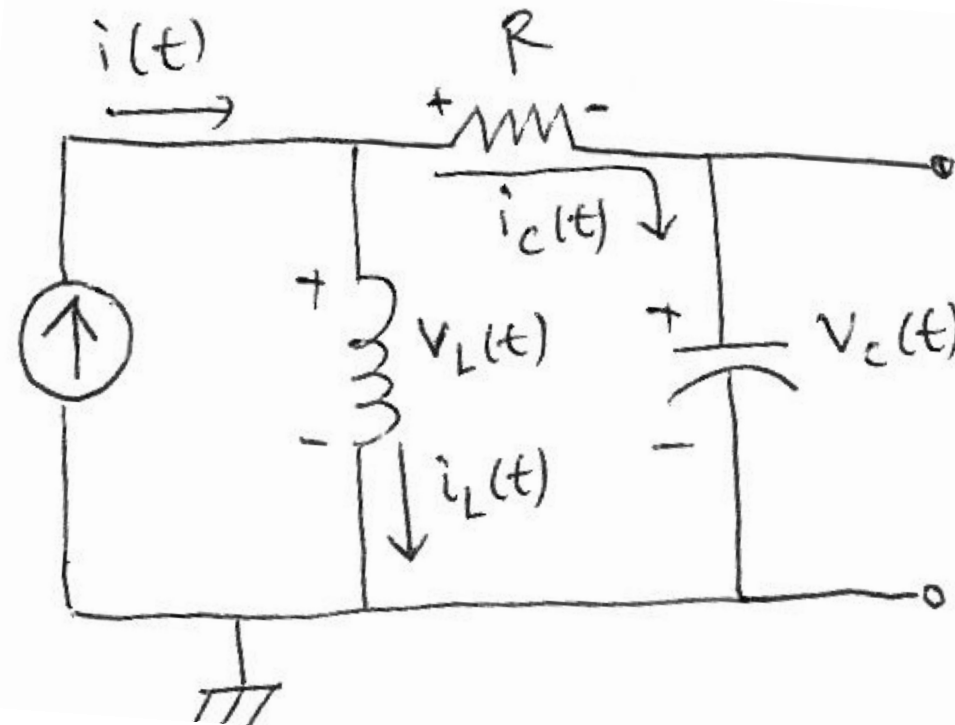
# Control canonical analog computer ( $m < n$ )

- Before the digital age continuous-time continuous-amplitude analog computers made of analog electronic circuits were considered. State space models can be realised as analog computers.



# Example – electrical circuit

- Consider the RLC circuit



- The input is the current  $i(t)$  and the output is the voltage  $v_C(t)$ . Thus,  $u(t) = i(t)$  and  $y(t) = v_C(t)$ .

# Example – electrical circuit

- The RLC circuit has transfer function

$$H(s) = \frac{V_C(s)}{I(s)} = \frac{\frac{1}{C}s}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

and control canonical representation for  $H(s)$  is:

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -\frac{1}{LC} & -\frac{R}{L} \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$
$$y(t) = \begin{bmatrix} 0 & \frac{1}{C} \end{bmatrix} x(t)$$

# Modal canonical form

- This is state-space representation based on partial fraction expansion.
- Assume that transfer function  $Y(s)/U(s)$  has real distinct poles  $p_1, p_2, \dots, p_n$ , and take  $m < n$ . In this case, we can write:

$$\frac{Y(s)}{U(s)} = \frac{c_1}{s - p_1} + \frac{c_2}{s - p_2} + \dots + \frac{c_n}{s - p_n}$$

- Each component (mode) of expansion on the right has an easy state space representation.

# Modal canonical form

- The component

$$H_k(s) = \frac{c_k}{s - p_k}$$

can be realized with the one state system

$$\dot{x}_k(t) = p_k x_k(t) + u(t)$$

$$y_k(t) = c_k x_k(t)$$

Taking Laplace transforms of  $u(t)$ ,  $x_k(t)$ , and  $y_k(t)$  shows that this system has transfer function  $H_k(s)$ .



# Modal canonical form

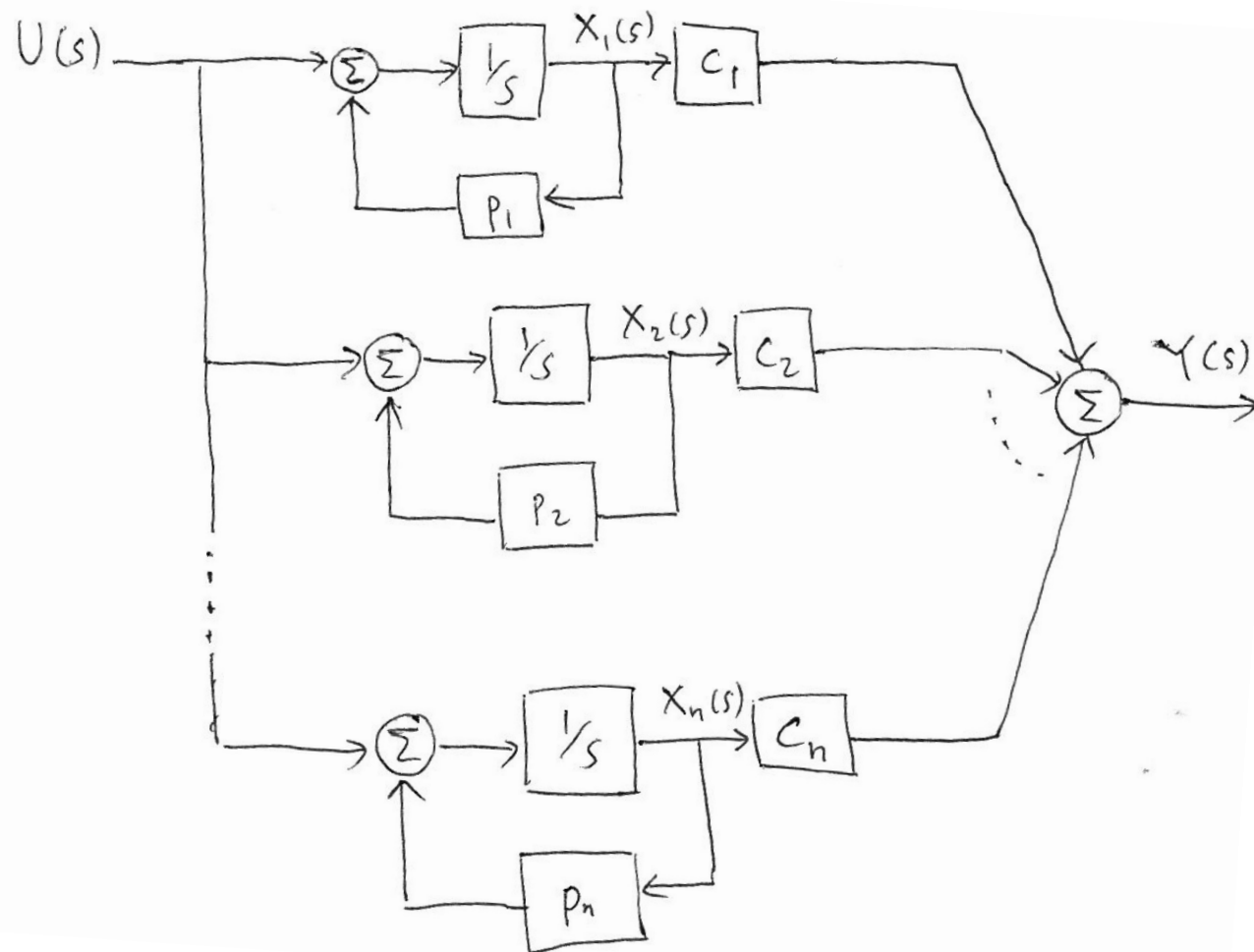
- When all poles real and distinct we have the modal canonical state-space representation (with  $x(t) = (x_1(t), x_2(t), \dots, x_n(t))^T$ )

Modal canonical form:  
 **$A$  matrix is real and diagonal**

$$\dot{x}(t) = \begin{bmatrix} p_1 & 0 & 0 & \dots & 0 \\ 0 & p_2 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & p_n \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \\ 1 \end{bmatrix} u(t)$$
$$y(t) = [c_1 \quad c_2 \quad \dots \quad c_n] x(t)$$

# Modal canonical form block diagram

- Block diagram



# Example – electrical circuit

- Referring again to previous RLC circuit example, suppose that  $R > 0$  and

$$\frac{R^2}{L^2} > \frac{4}{LC}$$

then transfer function  $H(s)$  will have two real and distinct poles, and we can write

$$H(s) = \frac{\frac{1}{2C} \left( \frac{R}{L} + \sqrt{\frac{R^2}{L^2} - \frac{4}{LC}} \right)}{s + \frac{1}{2} \left( \frac{R}{L} + \sqrt{\frac{R^2}{L^2} - \frac{4}{LC}} \right)} + \frac{-\frac{1}{2C} \left( \frac{R}{L} - \sqrt{\frac{R^2}{L^2} - \frac{4}{LC}} \right)}{s + \frac{1}{2} \left( \frac{R}{L} - \sqrt{\frac{R^2}{L^2} - \frac{4}{LC}} \right)}$$

# Example – electrical circuit

- Modal canonical form for circuit is

$$\dot{x}(t) = \begin{bmatrix} -\frac{1}{2}\left(\frac{R}{L} + \sqrt{\frac{R^2}{L^2} - \frac{4}{LC}}\right) & 0 \\ 0 & -\frac{1}{2}\left(\frac{R}{L} - \sqrt{\frac{R^2}{L^2} - \frac{4}{LC}}\right) \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(t)$$
$$y(t) = \begin{bmatrix} \frac{1}{2C} \frac{\left(\frac{R}{L} + \sqrt{\frac{R^2}{L^2} - \frac{4}{LC}}\right)}{\sqrt{\frac{R^2}{L^2} - \frac{4}{LC}}} & -\frac{1}{2C} \frac{\left(\frac{R}{L} - \sqrt{\frac{R^2}{L^2} - \frac{4}{LC}}\right)}{\sqrt{\frac{R^2}{L^2} - \frac{4}{LC}}} \end{bmatrix} x(t)$$

# Observer canonical form

- From transfer function we can write

$$s^n Y(s) + a_{n-1} s^{n-1} Y(s) + \dots + a_0 Y(s) = b_{n-1} s^{n-1} U(s) + b_{n-2} s^{n-2} U(s) + \dots + b_0 U(s)$$



$$Y(s) + a_{n-1} \frac{Y(s)}{s} + \dots + a_0 \frac{Y(s)}{s^n} = b_{n-1} \frac{U(s)}{s} + b_{n-2} \frac{U(s)}{s^2} + \dots + \frac{b_0}{s^n} U(s)$$



$$Y(s) = -a_{n-1} \frac{Y(s)}{s} - \dots - a_0 \frac{Y(s)}{s^n} + b_{n-1} \frac{U(s)}{s} + b_{n-2} \frac{U(s)}{s^2} + \dots + \frac{b_0}{s^n} U(s)$$

# Observer canonical form

- Continuing

$$\begin{aligned}
 Y(s) &= -a_{n-1} \frac{Y(s)}{s} - \dots - a_0 \frac{Y(s)}{s^n} + b_{n-1} \frac{U(s)}{s} + b_{n-2} \frac{U(s)}{s^2} + \dots + \frac{b_0}{s^n} U(s) \\
 &= \frac{1}{s} \left\{ -a_{n-1} Y(s) + b_{n-1} U(s) + \frac{1}{s} \left\{ -a_{n-2} Y(s) + b_{n-2} U(s) + \frac{1}{s} \left\{ \dots \right. \right. \right. \\
 &\quad \left. \left. \left. + \frac{1}{s} \left\{ -a_0 Y(s) + b_0(s) \right\} \dots \right\} \right\} \right\}
 \end{aligned}$$

# Observer canonical form

- Introduce

$$X_1(s) = \frac{1}{s}(-a_0Y(s) + b_0U(s))$$

$$X_2(s) = \frac{1}{s}(-a_1Y(s) + b_1U(s) + X_1(s))$$

$$\vdots$$

$$X_{n-1}(s) = \frac{1}{s}(-a_{n-2}Y(s) + b_{n-2}U(s) + X_{n-2}(s))$$

$$X_n(s) = Y(s)$$

# Observer canonical form

- Then we have

$$sX_1(s) = -a_0X_n(s) + b_0U(s)$$

$$sX_2(s) = -a_1X_n(s) + b_1U(s) + X_1(s)$$

$$\vdots$$

$$sX_{n-1}(s) = -a_{n-2}X_n(s) + b_{n-2}U(s) + X_{n-2}(s)$$

$$sX_n(s) = sY(s)$$

$$= -a_{n-1}X_n(s) + b_{n-1}U(s) + X_{n-1}(s)$$



# Observer canonical form

- Let  $x_1(t), x_2(t), \dots, x_n(t)$  be inverse Laplace transform of  $X_1(s), X_2(s), \dots, X_n(s)$ , resp., with  $x_1(0) = x_2(0) = \dots = x_n(0) = 0$ .

- By inverse Laplace transform, equations equivalent to:

$$\dot{x}_1(t) = -a_0 x_n(t) + b_0 u(t)$$

$$\dot{x}_2(t) = -a_1 x_n(t) + b_1 u(t) + x_1(t)$$

$$\vdots$$

$$\dot{x}_{n-1}(t) = -a_{n-2} x_n(t) + b_{n-2} u(t) + x_{n-2}(t)$$

$$\dot{x}_n(t) = -a_{n-1} x_n(t) + b_{n-1} u(t) + x_{n-1}(t)$$

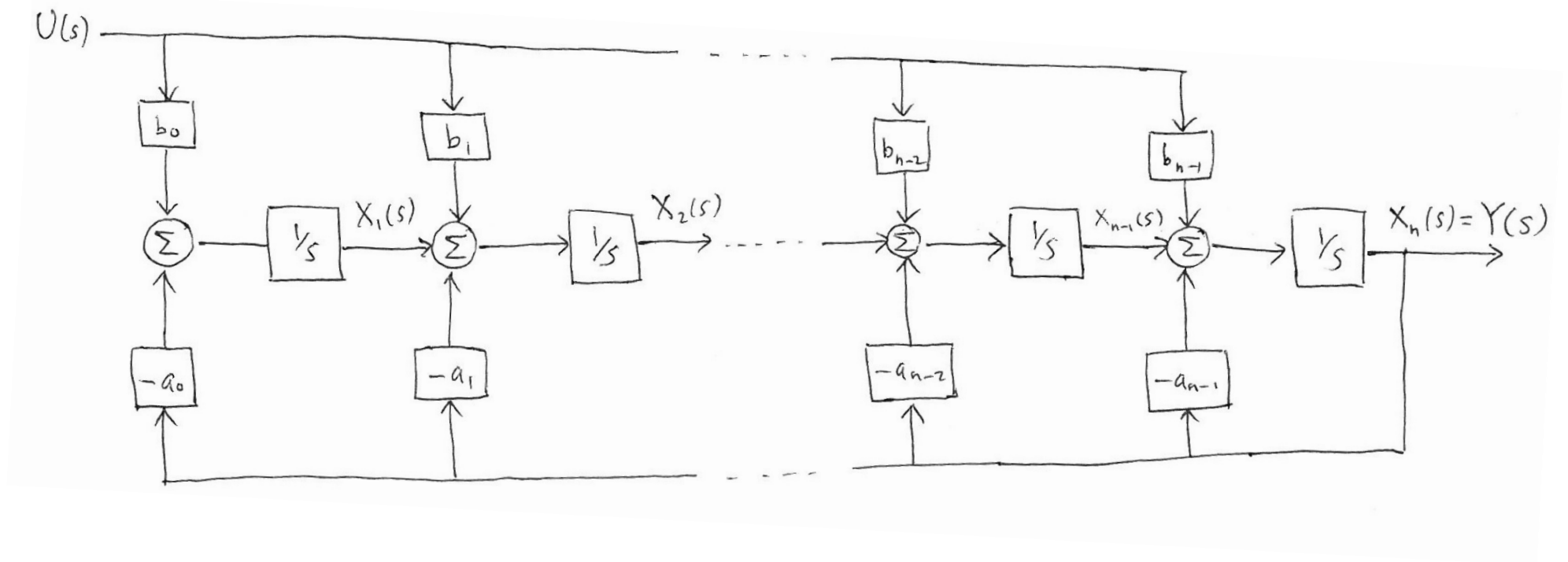
# Observer canonical form

- Letting  $x(t) = (x_1(t), x_2(t), \dots, x_n(t))^T$   
and collecting the equations together we have

$$\dot{x}(t) = \begin{bmatrix} 0 & 0 & 0 & \dots & -a_0 \\ 1 & 0 & 0 & \dots & -a_1 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & 1 & 0 & -a_{n-2} \\ 0 & \dots & 0 & 1 & -a_{n-1} \end{bmatrix} x(t) + \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_{n-2} \\ b_{n-1} \end{bmatrix} u(t)$$
$$y(t) = [0 \quad 0 \quad \dots \quad 0 \quad \dots \quad 0 \quad 1] x(t)$$

Observer canonical  
form

# Observer canonical block diagram ( $m < n$ )



# Example – electrical circuit

- For our favourite RLC circuit with transfer function

$$H(s) = \frac{\frac{1}{C}s}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

the observer canonical state-space representation is:

$$\dot{x}(t) = \begin{bmatrix} 0 & -\frac{1}{LC} \\ 1 & -\frac{R}{L} \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ \frac{1}{C} \end{bmatrix} u(t)$$
$$y(t) = \begin{bmatrix} 0 & 1 \end{bmatrix} x(t)$$

# Conversion from linear ODEs

- If you start with an ODE

$$y^{(n)}(t) + a_{n-1}y^{(n-1)}(t) + \dots + a_0y(t) = b_mu^{(m)}(t) + b_{m-1}u^{(m-1)}(t) + \dots + b_0u(t)$$

with non-zero initial conditions  $y(0)$ ,  $y^{(1)}(0)$ ,  $\dots$ ,  $y^{(n-1)}(0)$  then always use the observer canonical form, **not** the control canonical form.

- Why? For some instances of  $a_0, a_1, \dots, a_n$  and  $b_0, b_1, \dots, b_m$ , control canonical form cannot produce certain outputs  $y(t)$  with non-zero initial conditions.

# Shorthand notation

- Use  $S = (A, B, C, D)$  as shorthand for denoting the state-space representation/realisation

$$\dot{x}(t) = A x(t) + B u(t); x(0) = x_0$$

$$y(t) = C x(t) + D u(t)$$

- The matrices  $(A, B, C, D)$  are referred to as *system* or *state-space matrices*.

# State-space representation

- The state-space system

$$\dot{x}(t) = Ax(t) + Bu(t); x(0) = x_0$$

$$y(t) = Cx(t) + Du(t)$$

has explicit solution given by

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-s)}Bu(s)ds$$

$$y(t) = Ce^{At}x_0 + \int_0^t Ce^{A(t-s)}Bu(s)ds + Du(t)$$

# Matrix exponential

- Here  $e^{At}$  is a matrix exponential (Week 1). That is, it's the exponential of the matrix  $At$ .
- It is well-defined for every matrix  $A$  and every  $t$ .



# Matrix exponential

- Recall some properties

$$e^0 = I$$

$$e^{A(t+s)} = e^{At} e^{As}$$

$$e^{-At} = (e^{At})^{-1}$$

$$\frac{d}{dt} e^{At} = \lim_{h \rightarrow 0} \frac{e^{A(t+h)} - e^{At}}{h} = A e^{At} = e^{At} A$$

If  $A$  and  $B$  commute ( $AB = BA$ ) then  $e^{(A+B)t} = e^{At} e^{Bt}$

- Take care with the last property – not necessarily true if  $A$  and  $B$  do not commute (recall from Lecture 2)!

# Matrix exponential

- Given these properties of  $e^{At}$ , we can easily check that the given solution does indeed satisfy the ODE:

$$\begin{aligned}\frac{d}{dt}x(t) &= \frac{d}{dt}(e^{At}x_0 + e^{At} \int_0^t e^{-As}Bu(s)ds) \\ &= Ae^{At}x_0 + (Ae^{At}) \int_0^t e^{-As}u(s)ds + e^{At}(e^{-At}Bu(t)) \\ &= A\left(e^{At}x_0 + e^{At} \int_0^t e^{-As}u(s)ds\right) + (e^{At})(e^{At})^{-1}Bu(t) \\ &= Ax(t) + Bu(t)\end{aligned}$$

- Also, easy to see that  $x(0) = x_0$  (initial condition is satisfied).
- This solution is unique. That is, there can be no other solution of ODE other than solution above.

# Example - Solution of state-space system

- Consider state-space system

$$\dot{x}(t) = \begin{bmatrix} -5 & 18 \\ -3 & 10 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t); \quad x_0 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$y(t) = \begin{bmatrix} 1 & 1 \end{bmatrix} x(t) - 2u(t)$$

$$A = \begin{bmatrix} -5 & 18 \\ -3 & 10 \end{bmatrix} \quad \bullet \quad \text{Let } V = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

From Lecture 2, the matrix  $A$  has eigenvalues 1 and 4 and diagonalised by the matrix  $V$ .

# Example - Solution of state-space system

- Similar calculations to lecture 2 gives

$$e^{\begin{bmatrix} -5 & 18 \\ -3 & 10 \end{bmatrix} t} = V e^{\begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} t} V^{-1} = \begin{bmatrix} 3e^t - 2e^{4t} & -6e^t + 6e^{4t} \\ e^t - e^{4t} & -2e^t + 3e^{4t} \end{bmatrix}$$

- So,

$$x(t) = \begin{bmatrix} 2e^{4t} \\ e^{4t} \end{bmatrix} + \int_0^t \begin{bmatrix} (-6e^{t-s} + 6e^{4(t-s)})u(s) \\ (-2e^{t-s} + 3e^{4(t-s)})u(s) \end{bmatrix} ds$$

$$y(t) = 3e^{4t} + \int_0^t (-8e^{t-s} + 9e^{4(t-s)})u(s)ds - 2u(t)$$

- Going beyond these lecture notes for LTI state-space models (also available on the Course Outline):
  1. Franklin, Powell and Emami-Naeini, Feedback Control of Dynamic Systems, 6th edition, Addison-Wesley.
  2. Two courses EE263 “Introduction to Linear Dynamical Systems” and EE363 “Linear Dynamical Systems” at Stanford University (USA). All info available on the websites: <http://www.stanford.edu/class/ee263/> and <http://www.stanford.edu/class/ee363/>